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The Developer's Guide to Connecting CRM Data, AI and App Experience, at Scale

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The New Stack

The Developer's Guide to Connecting CRM Data, AI, and App Experience, at Scale

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Heroku is your fastest path from idea to production. Our fully managed application platform provides scalable app hosting so you can stay focused on building great apps.



Introduction

At the end of 2022, OpenAI released ChatGPT to the public. In the short term, generative AI has taken the world by storm: Some 700 million people use it every week, according to the company. And now the next phase is ongoing. Unlike generative AI's reactive content generation, agentic AI enables proactive, autonomous task execution and multi-step workflows.

According to [PwC's AI Agent Survey](#), 79% of senior executives say their companies have already adopted AI agents. Two-thirds of them report that their agents deliver measurable value by increasing productivity. Some 90% of executives plan to raise AI spending in the next 12 months.

Among the reported benefits of agentic AI systems, they note being able to:

- Perform tasks without constant human oversight
- Execute complex problem-solving tasks
- Respond in a nuanced and context-dependent way to queries
- Improve agent accuracy and customer satisfaction through a continuous feedback loop that helps AI steadily improve results

But the success of any agentic AI development project is dependent on the data that informs it. Without a foundation built on a comprehensive data platform, organizational leaders inevitably become hesitant to trust AI's conclusions. They question the reliability of outputs and are reluctant to turn over control of even basic tasks to autonomous AI agents. Hence, they fall

well short of achieving the goals set for AI deployments. The absence of the right data foundation makes it impossible for developers to design apps rapidly at scale and have the enterprise trust their conclusions in real time. This is a primary factor limiting the benefits obtainable from enterprise AI projects.

“Access to the CRM database alone is rarely sufficient for agentic AI success, as the business context is largely omitted.

Why are AI project failure rates so high?

It's no wonder so many AI development projects fail. Poor-quality data, a lack of organizational knowledge, and incorrect context stand in the way of success. That's why AI agents are prone to draw flawed conclusions. Their decisions are based on incomplete, biased, or stale datasets. It's a finding supported by [Gartner surveys](#). The IT analyst firm found that more than 40% of agentic AI projects will be canceled by the end of 2027 due to high costs, nebulous business value, and inadequate risk control.

It is becoming abundantly clear that siloed enterprise data must be integrated into AI apps and infrastructure as a vital element of DevOps workflows. By providing developers with easy access to complete data sets and the freedom to work with the languages and frameworks they prefer, they can build next-generation agentic AI applications that deliver tangible, measurable return on investment (ROI).

Access to the CRM database alone is rarely sufficient for agentic AI success, as the business context is largely omitted. AI agents need secure, real-time access to all enterprise data (whether structured or unstructured). This includes ERP, finance, production, inventory, supply chain management, business intelligence, product development, shipping, and more.

Yes, many agentic AI tools are customer-centric. They should be tightly coupled with CRM databases. However, they must also be able to instantly retrieve any relevant data wherever it resides — in enterprise databases, systems, or business logic — and piece together the intricate threads of enterprise context.

The vast troves of customer data within the Salesforce ecosystem, for example, as well as all business-critical data outside their CRM, must be easily accessible to any agentic AI tools being developed.

Accuracy versus latency

When it comes to incorporating unstructured and external data sources into AI models, there is often a latency-versus-accuracy trade-off. As more unstructured data is included in a model to achieve the right context, latency tends to suffer. If accuracy is already 95%, for example, a 1% increase could require a doubling of model complexity; however, that added complexity increases latency — response speed suffers.

Retrieval-augmented generation (RAG) has proven itself to be a way to resolve the trade-off between accuracy and latency. RAG automatically feeds current and relevant proprietary data into AI models. By supplementing them with trusted internal data sources, AI outputs improve, as fresher information from verified sources can be used. RAG makes it possible to incorporate both structured data from spreadsheets and relational databases and unstructured data from emails, PDFs, chats, social media, and more. It pre-processes and indexes this information, then uses semantic search tools to retrieve what's needed for a specific query by

referencing internal data pools and broader AI model data, providing more relevant answers.

The RAG methodology is built into Heroku's platform-as-a-service approach. It combines vector storage, AI model inference, and orchestration tools, enabling developers to build, deploy, and scale applications without managing complex infrastructure. In addition, it offers an abstraction layer for agentic AI tools. Heroku integrates closely with Salesforce to provide scalability and ease of development on any desired language or platform.

Beyond being a hosting or application platform, it is an environment where developers can build custom applications and logic that extend Salesforce's reach across the enterprise landscape. Developers can use these tools to offload complex compute, infrastructure management, and specialized development tasks, enabling them to focus on rapidly deploying sophisticated, context-aware applications with a lean operational footprint.

Those using Heroku cite the following benefits:

- Measurable economic return from AI projects
- Operationalization of decision authority
- Platform-controlled governance
- Closing the loop on the reasoning layer of agentic AI development
- Scaling promising agentic AI applications securely across the enterprise
- Eliminating much of the risk that haunts most agentic AI projects

By using Heroku as an AI abstraction layer, developers can harness whatever model and language they prefer without vendor lock-in. They can use it to link Salesforce CRM databases with enterprise data to produce AI agents that achieve their goals.

In this eBook, therefore, you will learn:

- The importance of a data foundation that extends beyond CRM to all the nooks and crannies of the enterprise.
- Why Heroku is a solid choice for anyone developing modern AI architecture.
- What Heroku is and how it works.
- How Heroku works within the Salesforce ecosystem to help developers create AI tools that operate on a comprehensive enterprise context to deliver measurable results.

CHAPTER 01

The importance of the data foundation

Most developers find it relatively straightforward to provide AI applications and agents with access to structured data in databases, Excel files, web forms, and other standardized systems. But that applies to only a fraction of the total knowledge base of any business. Between [80% to 90% of all enterprise data](#) is unstructured: social media posts, PDFs, manuals, onboarding docs, FAQs, text documents, customer interaction records, and emails. Without integrating this vital organizational know-how, agentic AI development projects often go astray. Their developers are forced to code without real-time access to a trusted and comprehensive data foundation. The agents they design that operate on incomplete data and missing context are not simply inefficient and unreliable; they represent a significant source of operational, financial, legal, and reputational risk.

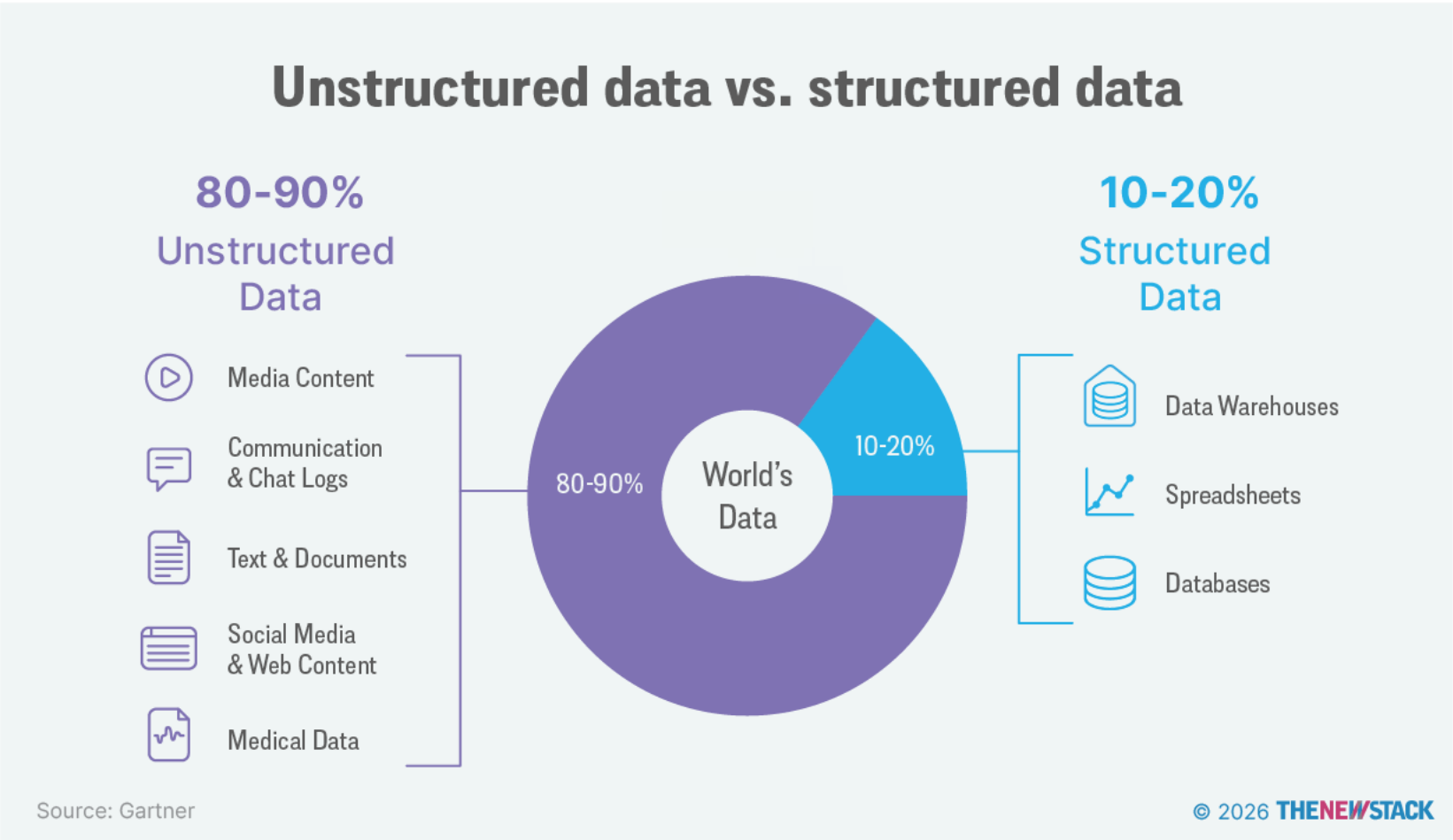


FIG 1.1 According to Gartner, at least 80% of enterprise data is unstructured, with some estimates going as high as 90%.

This developmental roadblock occurs because most organizations trap unstructured data in numerous information silos and data lakes. If the AI model and the AI agent can only interact with structured data and the public internet, they are missing crucial enterprise context. The agent cannot avoid reaching faulty conclusions because it cannot consider the full picture.

AI agents need access to diverse data sources across the entire organization.

This must include all areas of unstructured data. By unlocking the many silos that house enterprise knowledge, agents can glean previously unavailable context. Only by pooling data from policies, contracts, sales information, social media, email, and more can a unified, harmonized, and trusted data profile be created for each customer. Developers need that level of information availability if their AI projects are to achieve their goals and scale effectively.

How the lack of context limits AI agent effectiveness

Context, too, should extend to “agentic memory,” i.e., real-time contextual awareness within the AI agents themselves, so they remember previous interactions and respond relevantly. This is essential for both system connectivity and ongoing human-agent interactions.

Take the case of an AI agent aiding a human in a decision related to the raising of a credit limit. If the agent has access to only a limited range of data, an incorrect decision is likely – either denying extra credit to a valuable customer or extending credit to a high-risk individual. However, the chances of an agentic AI error diminish sharply when the agent can incorporate policy, all relevant customer information, the person’s credit history and credit rating, and their sales and marketing contact history into its computations.

Thus, developers must ensure their AI applications can access the full ecosystem of data that comprises customer and organizational context. This requires a real-time, scalable data

platform that provides unified, harmonized data, builds confidence in AI outcomes, and enables seamless integration with existing enterprise infrastructure.

“Trust becomes all the more essential as organizations attempt to scale promising AI systems across the enterprise.

Agent autonomy must be earned through trusted outcomes

In addition to gathering all available enterprise context, AI agents' autonomous actions must be trustworthy. They must respect all privacy and governance policies and act only on complete data.

Trust comprises three distinct aspects:

- Sufficient accuracy and relevance to eliminate errors, guesswork, and hallucinations.
- Responsible adherence to cybersecurity standards and privacy mandates through practices such as masking sensitive or personally identifiable information in real time, and automated tagging, classification, and policy enforcement.
- Comprehensive tracking of data lineage and provenance, which makes it possible to trace origin and isolate all changes, authorized or unauthorized, as a key

element of compliance with the EU's General Data Protection Regulation (GDPR) and other regulatory frameworks.

Trust becomes all the more essential as organizations attempt to scale promising AI systems across the enterprise. If the agents created by developers are to act autonomously across millions of transactions, the ramifications of mistakes or inaccuracies are magnified sharply. Consumer-oriented ChatGPT-based AI tools can perhaps tolerate some small degree of inaccuracy. But in an enterprise agentic AI setting, any shortcomings can be catastrophic. Agentic AI, then, not only raises the bar on trustworthiness but also takes it to a whole new level. When done correctly and based on the right data platform, the organization can steadily gain confidence in the actions taken by developer-designed autonomous agents.

Agentic AI must scale

Harmony, unification, and trustworthiness, though, cannot be achieved only in a small number of transactions or through laborious, time-consuming processes. They must be supported by speed, in real time, and at massive scale. Whether an organization has a thousand or hundreds of millions of customers, the data foundation available to developers must be robust and scalable enough to support agentic AI activity during moments when agents are fulfilling customer requests.

The underlying data platform, therefore, should seamlessly power both insights and actions with fluidity, distinguishing it from a passive information repository locked inside a storage array or data lake. A real-time, scalable foundation means data is available at the “right time for the right need,” ranging from sub-second needs for personalized web experiences to within-seconds delivery for customer service agents. For high-frequency interactions, the scalability of the underlying data platform becomes imperative.

Upon such a strong and scalable bedrock, developers can assemble agentic AI applications that can:

- Take consistent actions within defined bounds
- Deliver accuracy grounded in verifiable sources
- Provide sufficient control to ensure agents follow explicit instructions
- Safeguard the organization through guardrails that ensure security and avoid harm or malicious exploitation.

However, there is a gap in information visibility: [94%](#) of business leaders worldwide feel they aren’t taking full advantage of the information potentially available to them. They know, inherently, that the data is there to be found – somewhere. There is just too much fragmentation across silos for them to find it promptly.



FIG 1.2 Salesforce surveys indicate that the vast majority of business leaders are not taking advantage of their data.

Organizations often try to solve this problem by gathering a group of personnel in a room for hours or even days to collect and compile data to gain a more accurate picture of what is going on.

When AI tools are implemented, they often face a similar challenge – detecting or integrating them into workflows that handle the massive amount of unstructured enterprise

knowledge containing valuable insights. Why? Legacy data stacks typically relied on rigid ETL workflows interacting with on-premises infrastructure.

But a series of platforms has emerged, offering developers an abstraction layer that uses a modern cloud-based approach to data management. This helps to streamline ELT pipelines and offers the scalability, flexibility, and speed demanded by AI applications. These platforms include AWS Amplify AI kit for Amazon Bedrock, Vertex AI for Google Cloud, Azure OpenAI Service, and others. They are part of the modern data stack and can interact seamlessly with tools and applications such as Apache Kafka, Snowflake, and Databricks. This enables them to provide the level of modularity, efficiency, and AI readiness needed to support the latest generation of analytics and machine learning workloads.

Salesforce Data 360 provides a solid foundation

These challenges are also addressed by [Data 360](#), the only data engine native to Salesforce. It allows developers to activate trusted data everywhere across apps and agents. It powers workflows, decisions, and analytics with one complete view of the customer.

For example, it links CRM data to the millions of records in a warehouse system's product inventory. Every engagement, every workflow, every AI agent can be powered by the full depth of enterprise data. Data 360 bridges the gap between having the data and being able to act on it effectively in real time.

As it is deeply integrated with Salesforce and its metadata framework, Data 360 plays an essential role in establishing a trusted foundation of unified customer data that all Salesforce applications require. Its array of connectors provides easy access to business applications and tools, including SAP, Workday, AWS, Google, and Azure. This furnishes developers

with a platform that provides AI agents with the context they need to respond and act rapidly and with certainty.

“Every engagement, every workflow, every AI agent can be powered by the full depth of enterprise data.”

Key differences: Zero copy vs. traditional ETL		
Feature	Traditional ETL / Reverse ETL	Salesforce Zero Copy Integration
Data Movement	Physical transfer and replication (copying)	No movement; data stays at source
Data Latency	High (Batch updates, hours/days)	Real-time or near real-time
Implementation	Slow (Weeks or months to build pipelines)	5-10x faster implementation
Storage Costs	High (Duplicate storage in Data Cloud)	Low (No duplicate storage)
Maintenance	Brittle, high maintenance, error-prone	Stable; metadata-driven connections
Security	Risk of sensitive data exposure in transit	Data secured at source, governed by the original system

Source: Salesforce

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FIG 1.3 Key differences: Zero copy vs. traditional ETL

Data 360 accomplishes this without moving any enterprise data. Its Zero Copy feature, for instance, lets developers incorporate structured and unstructured data from any source, including data previously trapped in data lakes and warehouses, into Salesforce, where it can power automation, analytics, and customer personalization.

Zero Copy allows Salesforce to access data from Snowflake, Databricks,

Google BigQuery, and external data warehouses without physically copying, moving, or transforming it. Traditional ETL processes, on the other hand, entail complex pipelines for moving, restructuring, and storing data in Salesforce. Zero Copy bypasses this complexity while providing real-time access, reducing storage costs, and improving data governance.

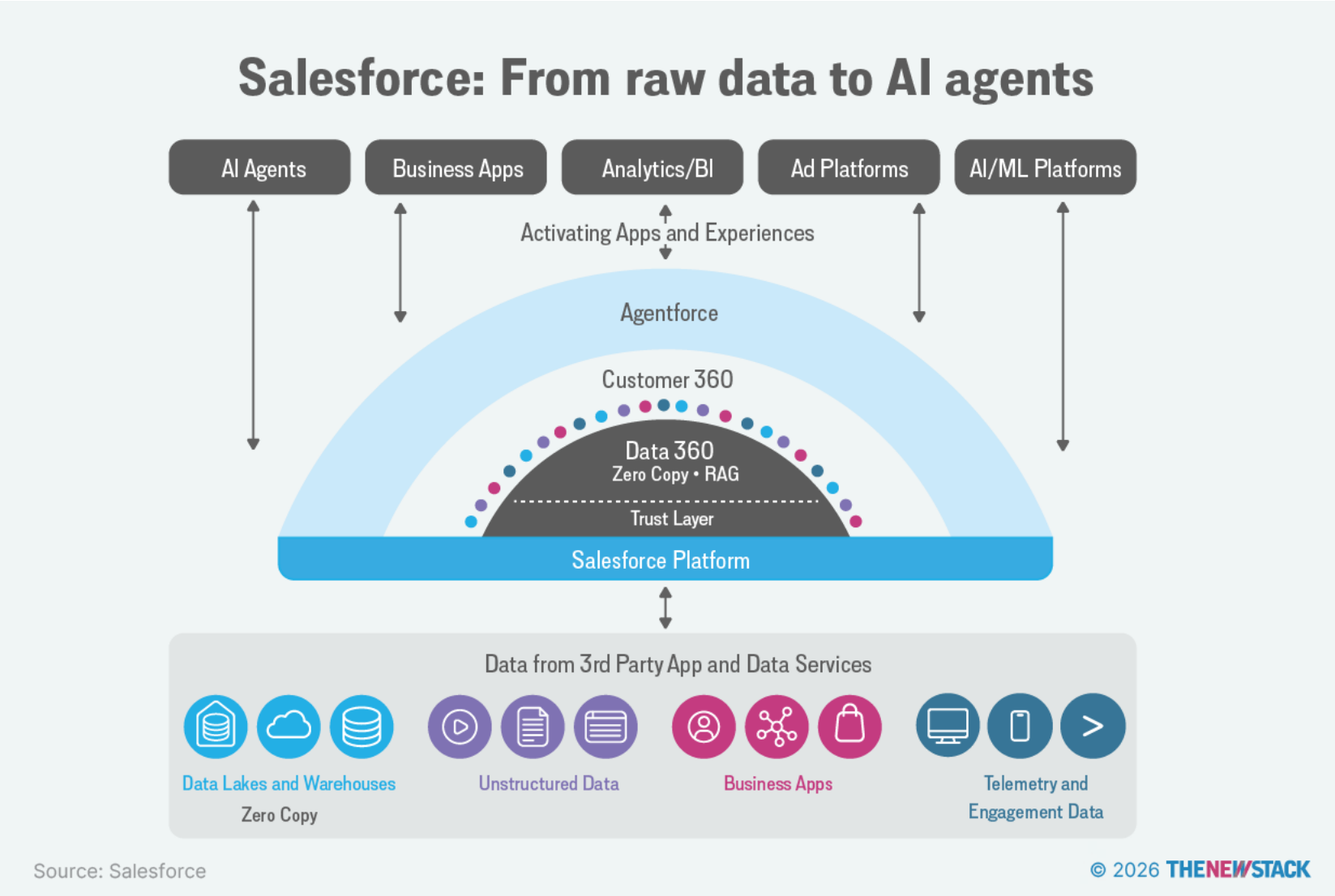


FIG 1.4 Data 360 maximizes the value of Salesforce CRM data while Agentforce adds the ability to compose AI agents that can operate on Salesforce database information to set up a wealth of autonomous tasks.

Data 360 simplifies data activation by unifying all information sources—from warehouses to lakes—and plugging them directly into the applications and workflows developers use. It possesses sufficient intelligence to process unstructured documents (such as reports and emails), transforming complex content into clear, structured insights that AI agents can use. By integrating unified knowledge directly with structured data, developers can easily build tools that scale to enterprise levels.

“By integrating unified knowledge directly with structured data, developers can easily build tools that scale to enterprise levels.

Agentforce adds agentic AI capabilities within Salesforce

Salesforce recently added agentic AI capabilities to its ecosystem.

Agentforce is a Salesforce platform where developers can build, deploy, and manage autonomous AI agents using low-code tools that can leverage existing Salesforce workflows and data. It enables development teams to manage the complete agent development lifecycle with tools for building, testing, deploying, managing, and orchestrating AI agents at scale that work alongside humans to handle tasks across sales, service, and marketing. It accomplishes this while adding the guardrails, governance, and controls required in enterprise operations. It distinguishes itself from chatbots by using LLMs to reason, plan, and execute complex workflows independently—such as resolving customer cases or qualifying leads—securely using company data via Data Cloud.

In real-world examples, Agentforce has achieved gains such as deflecting 30% of customer service cases and reducing resolution times for tech support issues

by 88%. Some of the key features and capabilities of Agentforce include:

- **Autonomous operation:** Unlike copilots that only suggest actions, Agentforce agents can independently take action, such as updating CRM records, sending emails, or scheduling meetings.
- **Built-in trust:** Uses the Einstein Trust Layer to ensure data security, privacy, and safety, preventing unauthorized data access.
- **Low-code/no-code builder:** Allows users to define, configure, and monitor agent behavior and topics without complex coding.
- **Integration capabilities:** Connects with existing Salesforce tools, including Apex code, Flow, and APIs to interact with both internal and external systems.

- **Data cloud integration:**
Enhances accuracy by connecting agents to real-time, unstructured, and structured data, as explained on the Salesforce website.
- **Sales:** Drafting personalized emails, creating closing plans, and providing real-time call guidance.
- **Service:** Handling customer inquiries, resolving service cases, and automating support processes.

Common use cases include:

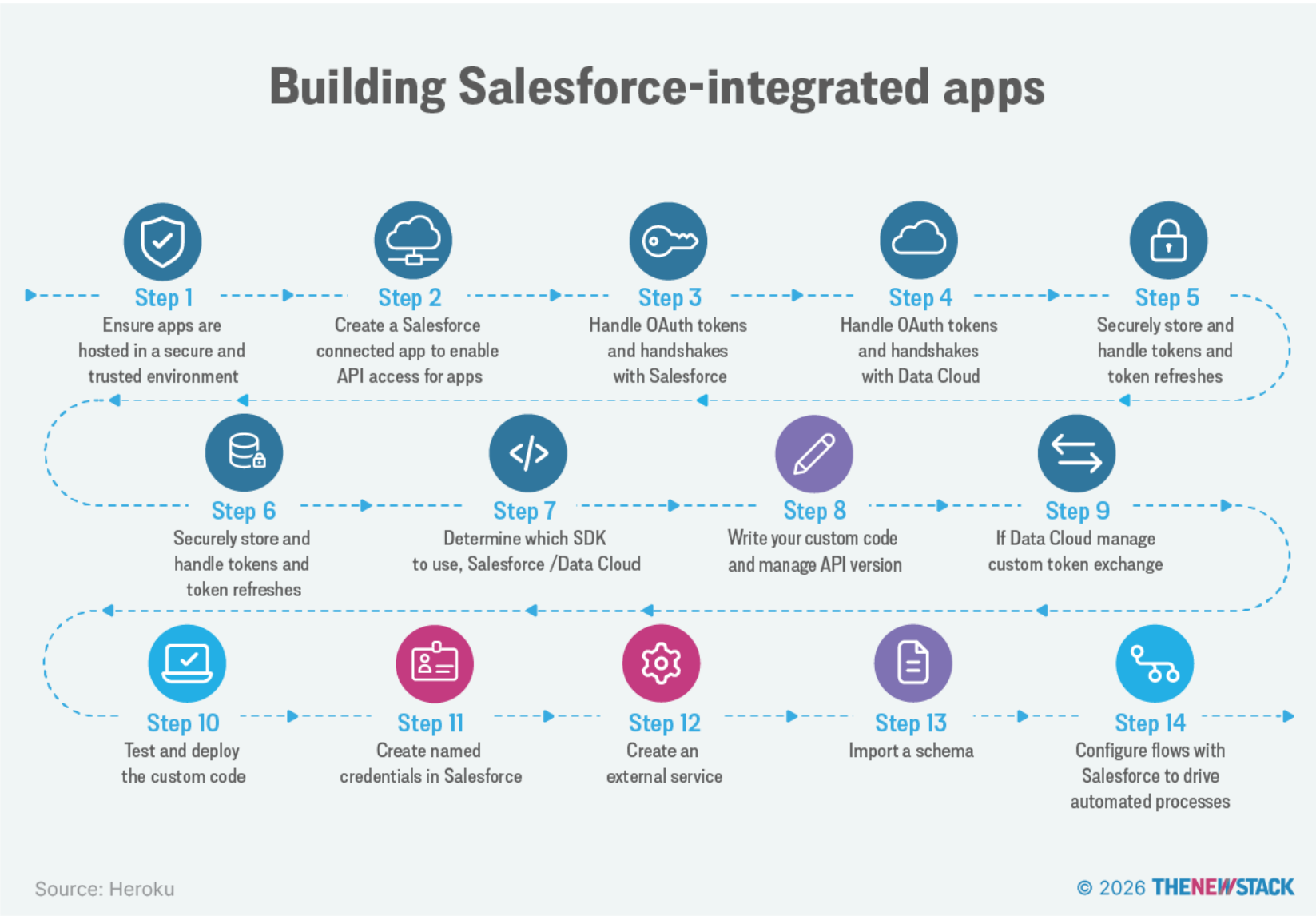


FIG 1.5 Due to the number of steps involved, building apps that integrate with Salesforce can be challenging.

- **Marketing & commerce:**

Automating campaign tasks and facilitating customer interactions.

In summary, Data 360 maximizes the value of Salesforce CRM data.

Agentforce adds the ability to compose AI agents that can operate on Salesforce database data to perform a wide range of autonomous tasks.

In the next chapter, we dig into how to extend those capabilities throughout the enterprise. By working with Heroku, developers can integrate the full range of enterprise data into the Salesforce ecosystem, greatly increasing the potential of agentic AI and AI apps across any organization.

“Unlike copilots that only suggest actions, Agentforce agents can independently take action

CHAPTER 02

Developers need a flexible platform to build customer-facing AI apps

The hype surrounding AI conveys the impression that AI is a wild success. Study after study suggests otherwise. Failure rates can reach 95%, according to [one MIT study](#). Notable AI and agentic AI failures over the past year or two include:

- Replit's coding assistant deleted an entire production database.
- The Grok chatbot provided detailed, accurate instructions on how to break into a politician's home in Minnesota in response to a user prompt.

Such incidents reveal the frailties of AI. Many of these failures stemmed from AI agents' inability to access vital information stored in organizational silos.

Another challenge sometimes encountered on the Salesforce platform is the number of steps required to build applications. Within Salesforce, it can add up to 14:

Step 1: Ensure apps are hosted in a secure and trusted environment.

Step 2: Create a Salesforce-connected app to enable API access to apps.

Step 3: Handle OAuth tokens and handshakes with Salesforce.

Step 4: Handle OAuth tokens and handshakes with Data 360.

Step 5: Securely store and handle tokens and token refreshes with Salesforce.

Step 6: Securely store and handle tokens and token refreshes with Data 360.

Step 7: Determine which SDK to use, Salesforce or Data 360.

Step 8: Write custom code and manage API version.

Step 9: Manage custom token exchange for Data 360.

Step 10: Test and deploy the custom code.

Step 11: Create named credentials in Salesforce.

Step 12: Create an external service.

Step 13: Import a schema.

Step 14: Configure flows with Salesforce to drive automated processes.

Heroku simplifies coding on Salesforce and enables developers to author agents that can access data anywhere in the enterprise, whether structured or unstructured. Instead of manually working through all the above steps,

Heroku provides automation and tools that focus on writing code rather than managing infrastructure.

These tools bridge the gap between open-source languages (Node.js, Ruby, Python, etc.) and the Salesforce data ecosystem. They eliminate complex API integrations to move data, integrate custom apps directly into Salesforce workflows in one click, enable secure, compliant synchronization of sensitive data, and add command-line access to Salesforce data. As a result, developers can more easily build customer-facing applications, custom dashboards, or data-intensive apps using their preferred technologies while leveraging the existing Salesforce CRM data.

Here are a few examples:

Kubernetes: Instead of wrestling with the configuration, scaling, and security complexities of setting up Kubernetes manually, Heroku abstracts away the operational overhead and enforces data segregation and access controls.

OAuth: Heroku addresses common challenges in OAuth integration, such as workflow complexity, secure communication, and token management, by allowing third-party apps, scripts, or services to access a Heroku account without sharing credentials. It supports web application authorization for third-party integrations and direct authorization for personal, non-expiring tokens.

Pinecone: Heroku simplifies Pinecone by acting as a streamlined hosting and integration platform. This allows developers to deploy AI-driven applications that use Pinecone's vector database without managing or maintaining infrastructure. This is accomplished by Heroku serving as an application layer that makes it easier to connect to, run, and scale AI services on Pinecone.

Python Quick Start Guide for `/v1/chat/completions` API

[Link](#)

```
import requests
import json
import os

# Global variables for API endpoint, authorization key, and model ID from Heroku config variables
ENV_VARS = {
    "INFERENCE_URL": None,
    "INFERENCE_KEY": None,
}

# Assert the existence of required environment variables, with helpful messages if they're missing.
for env_var in ENV_VARS.keys():
    value = os.environ.get(env_var)
    assert value is not None, (
        f"Environment variable '{env_var}' is missing. Set it using:\n"
        f"export {env_var}=$(heroku config:get -a $APP_NAME {env_var})"
    )
    ENV_VARS[env_var] = value

def parse_chat_output(response):
    """
    Parses and prints the API response for the chat completion request.

    Parameters:
        - response (requests.Response): The response object from the API call.
    """
    if response.status_code == 200:
        result = response.json()
        print("Chat Completion:", result["choices"][0]["message"]["content"])
    else:
        print(f"Request failed: {response.status_code}, {response.text}")

def generate_chat_completion(payload):
    """
    Generates a chat completion using the Stability AI Chat Model.

    Parameters:
        - payload (dict): dictionary containing parameters for the chat completion request

    Returns:
        - Prints the generated chat completion.
    """
    # Set headers using the global API key
    HEADERS = {
        "Authorization": f"Bearer {ENV_VARS['INFERENCE_KEY']}",
        "Content-Type": "application/json"
    }
    endpoint_url = ENV_VARS['INFERENCE_URL'] + "/v1/chat/completions"
    response = requests.post(endpoint_url, headers=HEADERS, data=json.dumps(payload))

    parse_chat_output(response=response)

# Example payload
payload = {
    "model": "claude-4-5-sonnet",
    "messages": [
        { "role": "user", "content": "Hello!" },
        { "role": "assistant", "content": "Hi there! How can I assist you today?" },
        { "role": "user", "content": "Why is Heroku so cool?" }
    ],
    "temperature": 0.5,
    "max_tokens": 100,
    "stream": False
}

# Generate a chat completion with the given payload
generate_chat_completion(payload)
```

Manual AWS-Salesforce integration versus using Heroku

Challenge	Manual AWS ↔ Salesforce Integration	Heroku ↔ Salesforce Integration
Architecture Complexity	Requires designing APIs, auth flows, middleware, and orchestration yourself	Heroku provides built-in patterns and services that simplify integration with Salesforce
Development Effort	Significant custom coding (REST/SOAP APIs, SDKs, event handling)	Leverages Heroku add-ons, prebuilt connectors, and Salesforce-native integrations to eliminate much of the manual labor
Authentication & Security	Must manually implement OAuth flows, token refresh, and secrets management	Heroku integrates smoothly with Salesforce identity and OAuth handling
Data Synchronization	Custom-built (polling, webhooks, queues via AWS services like Lambda/SQS)	Streamlined via Heroku Connect (bi-directional data sync with minimal setup)
Real-time Capabilities	Requires building event-driven systems (e.g., EventBridge, SNS)	Native support using Heroku + Salesforce Platform Events or Streaming API
Scalability	Scalable using AWS services like EC2, Lambda, S3, but requires tuning	Scales automatically via Heroku dynos with less infrastructure management
Maintenance Overhead	High overhead includes ongoing updates, monitoring, error handling, and retries	Low overhead as managed services reduce operational burden
Time to Market	Slower due to custom development	Faster due to prebuilt integrations and tooling
Cost Structure	Pay-per-use AWS pricing, but the engineering cost is higher	Heroku subscription + add-ons; may be more predictable, but sometimes higher per unit
Best For	Complex, highly customized enterprise architectures	Rapid development, SaaS apps tightly coupled with Salesforce
Typical Stack Components	Amazon Web Services services (Lambda, API Gateway, SQS, DynamoDB) + Salesforce APIs	Heroku apps, Heroku Connect, Postgres + Salesforce Platform

Source: Heroku

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FIG 2.1 Manual AWS-Salesforce integration versus using Heroku.

What is Heroku?

Heroku is a Salesforce-owned platform-as-a-service (PaaS) that runs on Amazon Web Services (AWS). It streamlines DevOps workflows, gives developers freedom to build with the languages and frameworks they prefer, and provides a suite of managed data services, including PostgreSQL.

The core use cases of Heroku include:

- Building high-traffic, customer-facing websites or mobile apps that require real-time Salesforce CRM data.
- Developers can use Heroku to connect any service, build agent actions using any programming language, and ground AI agents in the real-time context of key applications.
- Performing complex calculations, data crunching, or running AI models that exceed Salesforce's governor limits.
- Responding appropriately to Salesforce Platform Events.

Heroku integrates flexibly into the Salesforce ecosystem. Because it serves as an AI abstraction layer, developers do not need to manage the infrastructure underlying their applications. This saves valuable time and allows the focus to remain on app and agent creation. There is no need to worry about managing server and resource provisioning, networking, and security.

Heroku also integrates fully with development tools such as Datadog, GitHub, and Terraform to support modern CI/CD and Infrastructure as Code (IaC) workflows. For example, it provides first-party GitHub integration to connect applications directly to a GitHub repository.

How does Heroku work?

Heroku runs customer workloads in Linux containers called Dynos. The Dyno manager orchestration engine coordinates and monitors those containers. It detects software-level issues and addresses them. If an outage is caused by a malfunction in the underlying hardware, for instance, Dyno manager automatically moves the affected workloads to a new server.

In addition to Dynos, Heroku utilizes high-level application building blocks. One of the most popular is Heroku Postgres, a managed relational database. It automates infrastructure maintenance chores and performs a range of other tasks, such as:

- Periodically backing up its contents
- Enabling developers to quickly create copies of a database for testing and load balancing purposes.
- Support for an open-source tool called pgvector that enables it to store

embeddings, the mathematical objects in which AI models keep information. It enables developers to build full retrieval-augmented generation (RAG) applications or to leverage vector similarity search. As well as seamlessly storing and querying embeddings alongside app data using the same trusted Heroku database experience.

How Heroku aligns with Salesforce

Heroku ties into the Salesforce portfolio through a suite of native integrations within the Salesforce ecosystem. This makes it easier to deploy customer apps and apply AI agents in production environments while extending Salesforce functionality across the enterprise. Developers can use open-source languages such as Python, Ruby, Node.js, Java, Scala, and PHP while building customer-facing apps.

Heroku's cloud application hosting platform rapidly delivers scalable apps by transforming organizations into app companies. How? By allowing developers to spend their time developing and deploying apps that generate value. It makes the processes of deploying, configuring, scaling, tuning, and managing apps more straightforward, so developers can focus on what's important: building apps that engage customers. Instead of apps being detained in test and dev due to the inability to validate and scale them, Heroku gets good apps out into the wild fast, so customers can interact with them while minimizing risk.

The Heroku platform focuses on apps and the developer experience. It lets companies of all sizes embrace the value of apps, not the distraction of hardware and servers – virtual or otherwise. Further, it aids developers by offering immediate provisioning of a scalable, highly available database with rollback support for their apps and aligning with their development style. It achieves this by understanding that data lies at the heart of any significant app — whether it's customer data or data about the service it provides — an app and its data go hand in hand.

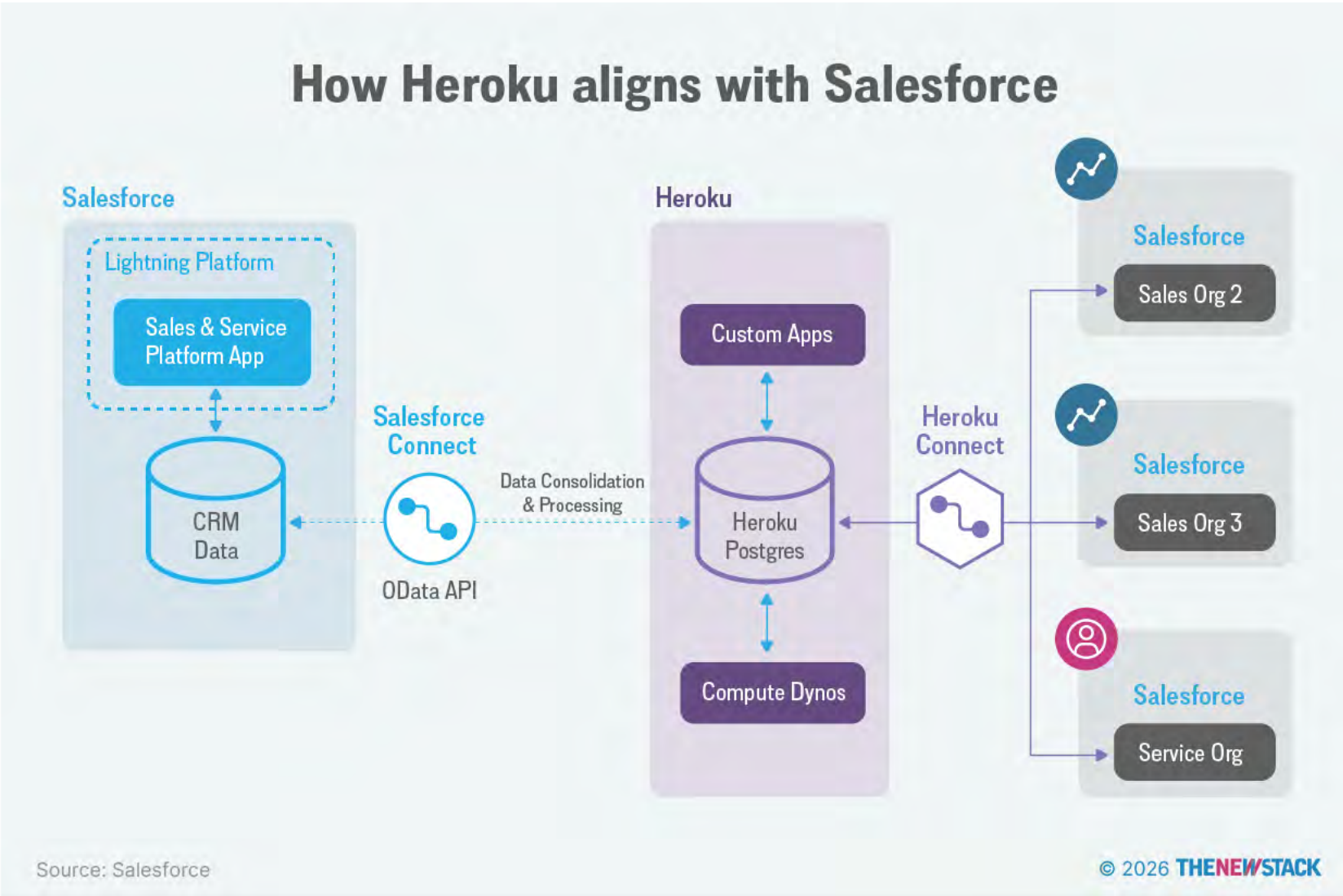


FIG 2.2 Heroku aligns seamlessly across the Salesforce ecosystem.

Heroku managed data services speed the path from prototype to production

Heroku is flanked by a series of managed data services and add-ons, including Heroku Connect, Heroku AppLink, and Heroku Managed Inference and Agents.

Heroku Connect: Synchronizes Salesforce data with a Heroku Postgres database, enabling bi-directional data flow. As the data bridge between

Salesforce and Heroku, any changes made in the Heroku database are automatically pushed back to Salesforce (and vice versa). In addition, Salesforce Objects (e.g., Accounts, Contacts, Custom Objects) are synced directly to Heroku Postgres tables.

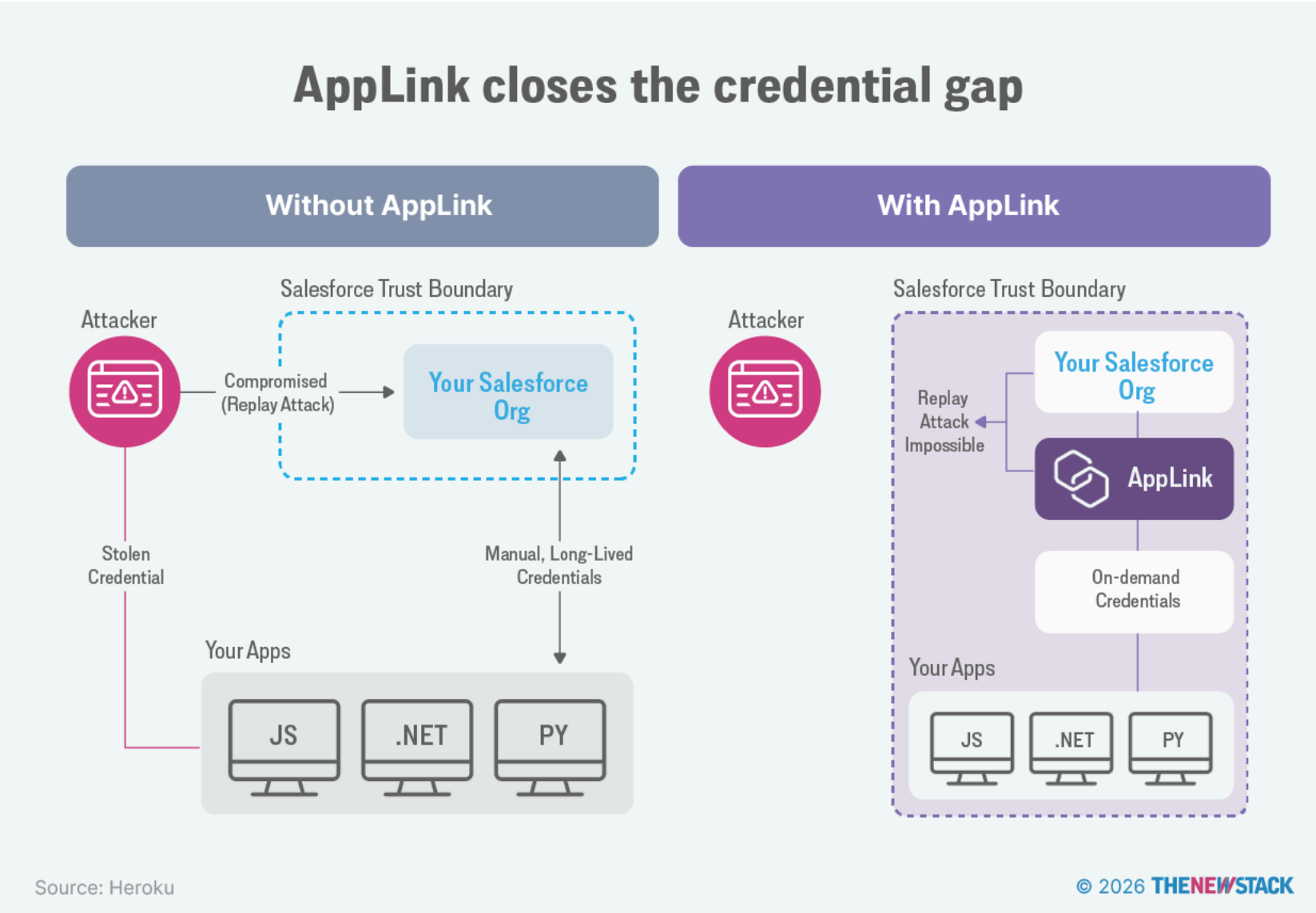


FIG 2.3 Heroku AppLink establishes a trust boundary around Salesforce and external applications.

Heroku AppLink: AppLink is the logic bridge between Heroku and Salesforce. It functions as a secure API gateway that exposes Heroku-hosted code as native “actions” inside Salesforce. It securely connects custom apps to Salesforce, allowing Salesforce Flow or Agentforce to trigger complex processes. Authentication is managed using short-lived tokens. This approach eliminates the need for manual OAuth management or “forever keys.” AppLink securely handles large attachments or files by processing them on Heroku and saving the results back to Salesforce

Files. In addition, it can pull data from multiple Salesforce organizations. If an enterprise comprises a complex, multi-organizational structure, AppLink can connect to all of them and pull Salesforce data from each.

Heroku Managed Inference and Agents: Developers use the Managed Inference and Agents Add-on to create custom AI agents. This opens the door to running the developer’s own LLM or selecting a preferred LLM, such as Anthropic’s Claude, GPT, or Grok. By using a Managed Inference API rather

than exposing raw GPUs on traditional dynos, developers can access preconfigured AI models (such as Claude Sonnet 4.6, Stable Image Ultra, or Qwen3 235B) via Heroku Managed Inference and Agents without the high, constant overhead of dedicated GPU infrastructure.

Model Context Protocol (MCP) on Heroku: Extends AI agent capabilities with tools and APIs using the open Model Context Protocol. Easily host and scale MCP servers on Heroku to

enhance agents with a broad ecosystem of tools or custom logic.

Now that we have covered what Heroku is, it is time to highlight how it amplifies the results developers can achieve on the Salesforce platform when designing custom AI apps and agents.

Execute Tools Automatically with **/v1/agents/heroku**

Example **/v1/agents/heroku** Request with mcp Tool

```
curl "$INFERENCE_URL/v1/agents/heroku" \
  -H "Content-Type: application/json" \
  -H "Authorization: Bearer $INFERENCE_KEY" \
  -H "X-Forwarded-Proto: https" \
  -d @- <<EOF
{
  "model": "claude-4-sonnet",
  "messages": [
    {
      "role": "user",
      "content": "Use the code exec tool to take the sha256 of the following string: 'Heroku Rocks'."
    }
  ],
  "tools": [
    {
      "type": "mcp",
      "name": "code_exec_python"
    }
  ]
}
EOF
```

CHAPTER 03

Bridging the gap between CRM data, app development, and agentic AI

Salesforce CRM databases contain a large portion of the world's critical customer data. This information drives the development of many customer-facing applications and agentic AI deployments. But beyond the CRM perimeter lies massive amounts of business-critical data that must be part of any holistic AI solution. Developers need to tap into all this data, along with any custom logic outside the CRM, to build context-aware AI applications and agents.

Heroku serves as a bridge among CRM, enterprise data, and application development. It prioritizes an open approach over rigid ecosystems, enabling developers to future-proof their AI stack as technology evolves. Using Heroku, they can leverage a secure, managed infrastructure that simplifies compliance and data safety, allowing them to focus on building features rather than navigating complex security checklists.

In tandem with Salesforce, it helps developers close the gap between prototype and production by providing a reasoning layer that enables agentic AI to move beyond conversation and into secure, scalable execution. In addition, it solves complex integration and data challenges across the Salesforce ecosystem, from Agentforce to Data Cloud, by giving developers the freedom to build with thousands of open-source libraries, add-ons, and frameworks.

Heroku extends the capabilities of Salesforce

To bring the most ambitious ideas to life, developers need creative flexibility. Heroku shifts the game from manual infrastructure management to shipping features, allowing lean teams to deliver AI without the burden of excessive operational overhead.

Remember the 14 steps required to build AI apps fully integrated with Salesforce? Heroku apps, building blocks, and automation help developers move from Step 1 to Step 14 without the intervening steps.

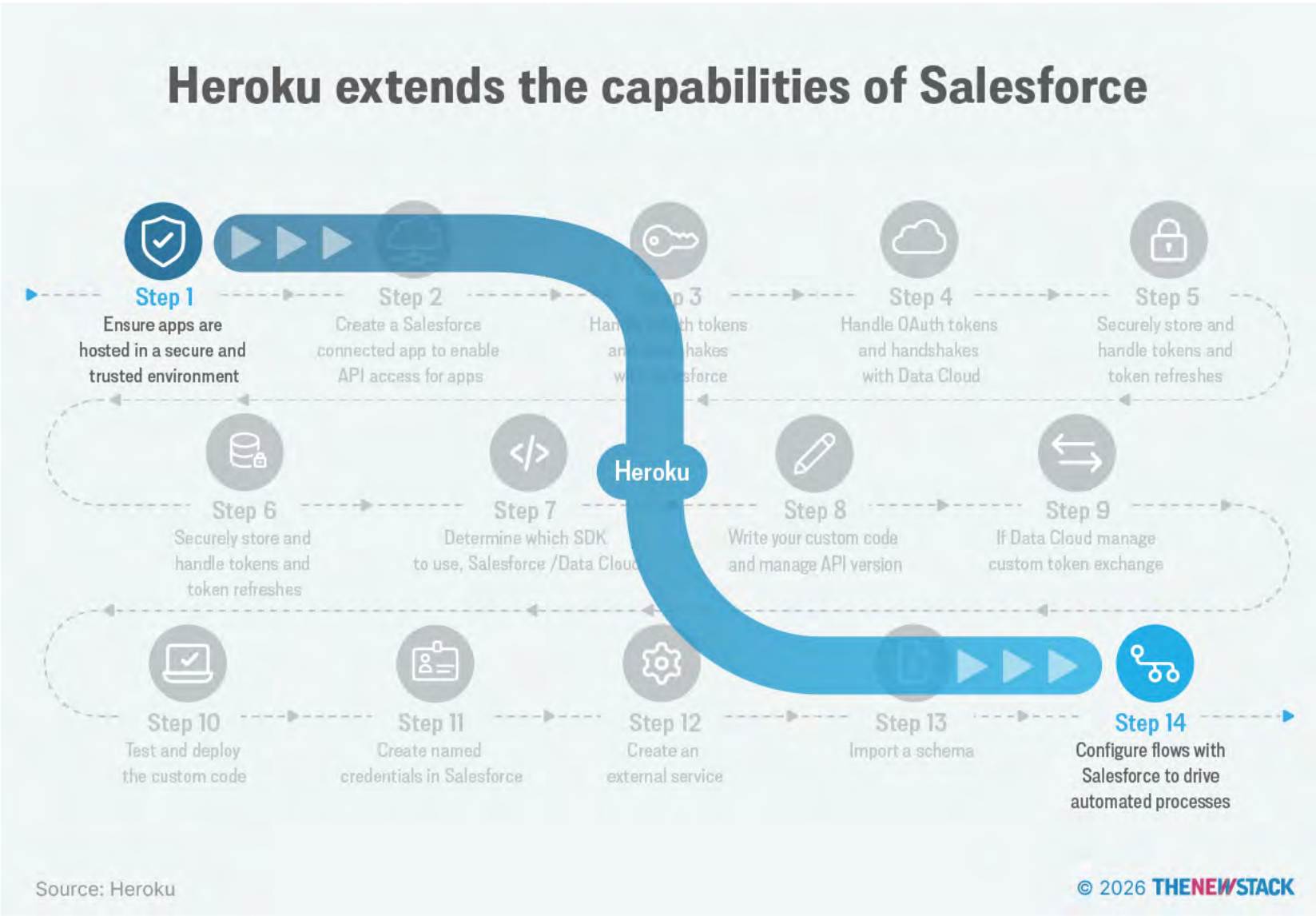


FIG 3.1 Heroku simplifies the integration path between Salesforce and external services by reducing infrastructure and workflow overhead.

Heroku tools, such as [Heroku AppLink](#) and [Heroku Connect](#), streamline integration, enabling developers to safely trigger custom logic and seamlessly synchronize data for a unified experience. By exposing all Heroku apps in Salesforce as services, developers can use AppLink to invoke Heroku applications through Salesforce Data 360 and Agentforce.

How about document generation? AppLink can bypass Apex heap limits to generate 50MB+ PDFs, complex Excel files, or high-res images on the fly. Similarly, its bulk processing capabilities can handle “one-off” heavy lifts (e.g., recalculating 100k records) that would time out in native Salesforce. When running AI models, performing complex math, or dealing with data transformations that exceed Salesforce governor limits, AppLink has become a go-to tool for developers.

Heroku Connect is another useful developer tool, as everything is handled within a trusted boundary rather than relying on external

services. Heroku manages all aspects of authentication to ensure that Salesforce and Heroku interact with a firm handshake and a secure token exchange. Further features include no-code mapping for point-and-click configuration – no need for custom API code or synchronization logic. And for apps that require fast, local SQL access to large volumes of CRM records, Heroku offers high-volume performance, bypassing Salesforce API limits. Heroku Connect makes it possible to:

- Design high-traffic web or mobile apps built on Node.js/ Python that need instant CRM data access.
- Develop standard SQL tools to query Salesforce data for complex analytics and reporting.

When it comes to security and compliance, Heroku adds another layer of protection, keeping Salesforce CRM data secure and

compliant at the infrastructure level. It offers [Private Spaces](#) for complete network isolation, while [Heroku Shield](#) provides the advanced controls needed to meet complex compliance requirements like HIPAA and PCI.

These are some of the capabilities of the Heroku platform. Let’s explore how Heroku’s and Agentforce’s agentic AI capabilities differ.

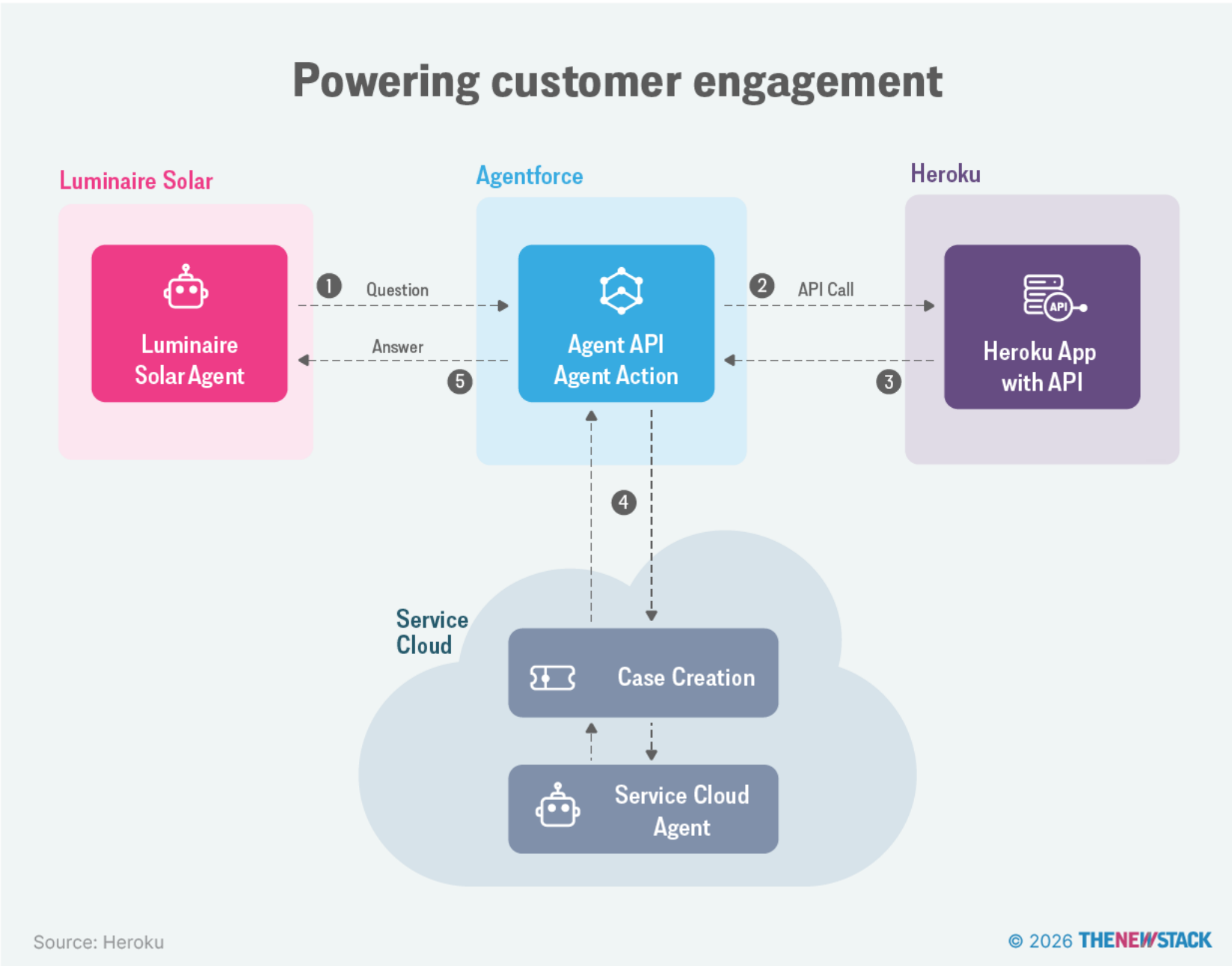


FIG 3.2 Agentforce works in conjunction with Heroku, which interacts with the Salesforce Service Cloud (bottom) and external AI agents (on the left).

CHAPTER 04

How Heroku works with Agentforce

There can sometimes be confusion between Agentforce and Heroku. It is understandable as both deal with AI agents. Let's cover what Agentforce is, compare it to Heroku, and then lay out how they complement each other.

As covered in [Chapter 1](#), Agentforce is a Salesforce platform where developers can build, deploy, and manage autonomous AI agents using low-code tools that can leverage existing Salesforce workflows and data. It enables development teams to manage the agent development lifecycle using a set of tools to build, test, deploy, manage, and orchestrate AI agents at scale. It accomplishes this while adding the guardrails, governance, and controls required in enterprise operations. It distinguishes itself from chatbots by using LLMs to reason, plan, and execute complex workflows independently—such as resolving customer cases or qualifying leads — securely using company data via Data Cloud.

The value of Agentforce

Agentforce helps organizations to scale their operations by allowing AI to handle repetitive or complex tasks, freeing employees for higher-value work. Included with it are: a web coding interface for generating app structures, custom objects, and code using AI; the

ability to create and configure autonomous agents using clicks to act as functional parts of an app; and a way to create intelligent, tailored mobile experiences while ensuring all generated applications follow compliance standards.

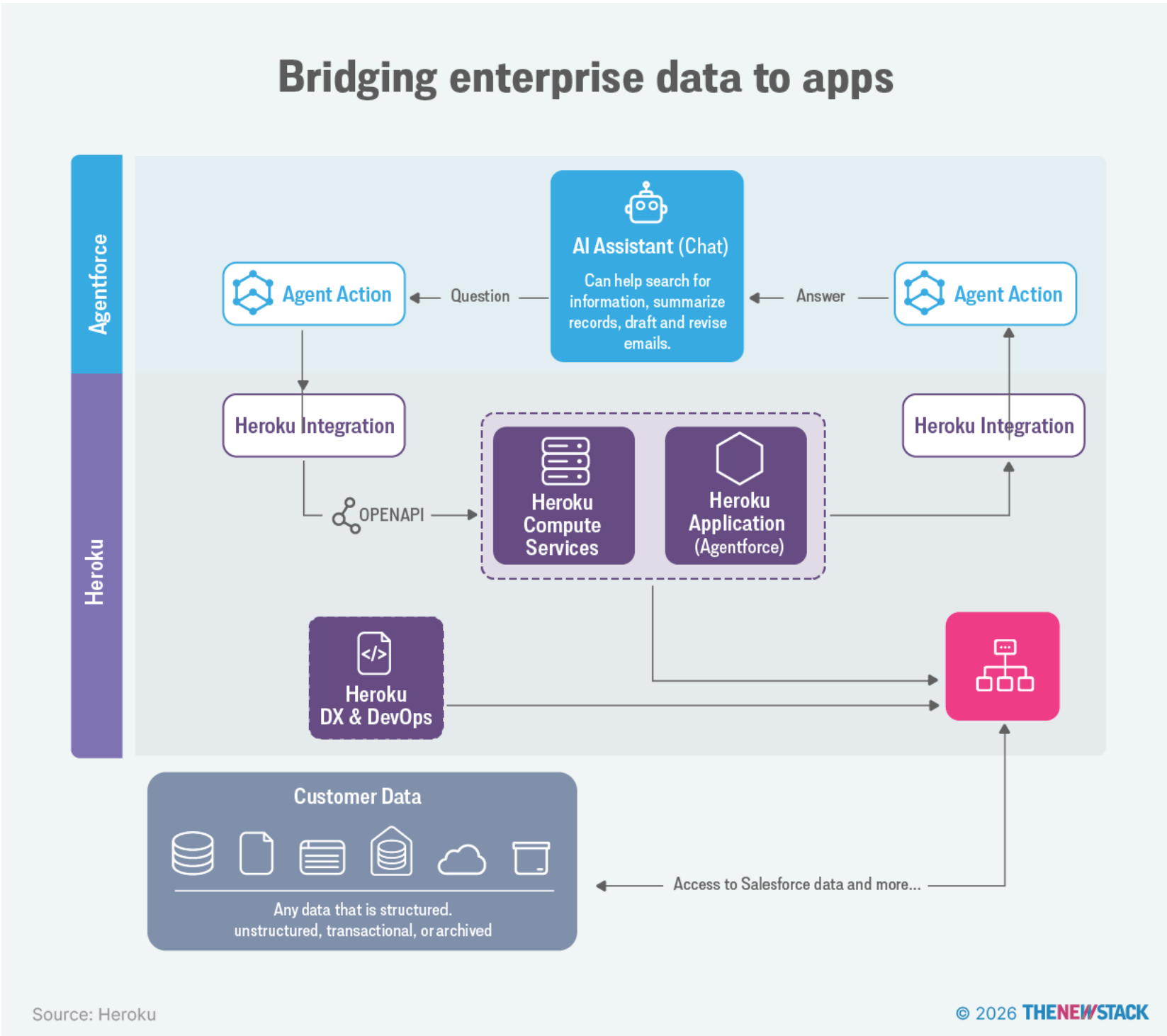


FIG 4.1 Heroku is the bridge between CRM, enterprise data, and application development.

Using conversational AI, Agentforce helps create project structures, generate code, and build web components. It accelerates app creation, manages data integration, and ensures secure deployment. Agentforce acts as an intelligent assistant throughout the development lifecycle, from planning and code generation to testing and deployment

“Using conversational AI, Agentforce helps create project structures, generate code, and build web components.”



SUMMARY

Heroku is an AI PaaS (Platform as a Service) for deploying and running modern apps, based on a managed container system, with integrated data services and a powerful ecosystem. The Heroku developer experience is an app-centric approach for software delivery, integrated with today's most popular developer tools and workflows. If agentic AI is to achieve success and scale across the enterprise, it must have at its disposal in real time all CRM and other enterprise data (whether structured or unstructured). Heroku performs this vital function while integrating closely with Salesforce to provide scale and ease of development in whatever language or platform is desired.

KEY FEATURES

- Build agentic AI applications in your favorite language that scale automatically
- Streamline app development & deployment and reduce operational complexity
- Integrate with Salesforce securely and easily

How does Agentforce align with Heroku?

Agentforce [Testing Center](#), for example, lets development teams run scenario-based, dataset-driven evaluations (including synthetic test cases) in a sandbox before go-live so that they can surface instruction-following and tool-use failures early.

Data 360's tight integration with the Salesforce Agentforce platform delivers structured and unstructured data from over 200 sources in real time. Once ingested, data is transformed and indexed, creating a unified foundation for Agentforce. This data is directly accessible for prompt templates and flow automations. Further, it provides developers with a unified data foundation baked into Agentforce, enabling them to turn existing data into action without moving or changing it. This simplifies tasks such as creating connected customer profiles across marketing, sales, service, and commerce, and activating them in real time across every channel.

To briefly summarize what we covered earlier:

Heroku acts as a bridge between CRM, enterprise data, and application development. It allows developers to focus on building features rather than navigating complex infrastructure and security checklists. It solves integration and data challenges across the Salesforce ecosystem by giving developers the freedom to build with thousands of open-source libraries, add-ons, and frameworks. It lets developers use any language to build integrated, data-driven solutions.

Heroku's agentic qualities arise from additional services, including Heroku Managed Inference, Agents, and the Model Context Protocol (MCP).

Heroku Managed Inference and Agents brings together a set of primitives that make it easier for developers to build, scale, and operate AI-powered features and applications, without the heavy lifting of managing AI infrastructure. Managed Inference and Agents also

allows developers to use their own model in the applications and agents they build for the Salesforce platform.

Managed Inference and Agents provides ready-to-use access to a curated set of AI models, selected for their generative power and performance, and optimized for ease of use and efficacy across a variety of domains.

Heroku enables developers to build, run, and scale AI applications that can reason, take action, and interact directly with data within the Heroku ecosystem. They act as a bridge between LLMs and custom application code, enabling agents to execute complex tasks that require more than just text generation. By executing code generated by LLMs within a secure sandbox environment (courtesy of Dynos), a variety of new capabilities are unlocked, including running SQL queries against databases, processing files, and interacting with other Heroku features.

As a result, agents can connect to application data and backend services. Instead of relying solely on base-model training data, agents can provide context-aware responses. In addition, Heroku agents carry out multi-step actions, such as retrieving data, analyzing it, and acting on it. As a result, developers can use CLI commands to add AI capabilities rather than build AI infrastructure from scratch.

“Managed Inference and Agents also allows developers to use their own model in the applications and agents they build for the Salesforce platform.”

How Model Context Protocol (MCP) extends Agentforce

MCP extends Heroku Managed Inference and Agents with a set of primitives and operations, giving agents new capabilities. For example, developers can create agents that execute code in Heroku Dynos and call tools and application logic. MCP simplifies building agents and workflows by standardizing how developers provide context and integrate tools. They can host and scale MCP servers on Heroku to improve agents with a broad ecosystem of tools or custom logic. This makes it possible to expose application logic, APIs, or custom tools to agents such as Agentforce, Claude, or Cursor via custom code.

In summary, Heroku Managed Inference and Agents and MCP make it easy to integrate with Salesforce Agentforce and extend its capabilities, enabling AI to perform custom actions, such as reading PDF documents or generating dynamic content.

For example, Agentforce automatically determines which Heroku application to

invoke for specific actions, such as compiling a report into a PDF from Salesforce data. Without Heroku, this task can be challenging due to the many steps involved – 14 in all, including authentication. Heroku AppLink bypasses this complexity by seamlessly exposing an external service to the developer. A Salesforce agent within Agentforce can then invoke that app and accomplish whatever is needed. Once published, actions can be called directly by Salesforce Flow, Apex, and Agentforce agents.

Within the Data 360 platform and Agentforce, developers can use Heroku to extend the reach of Salesforce assets across the enterprise. They can bridge CRM data with the flexibility of Heroku's ecosystem of managed data and AI services to build, deploy, and scale custom context-aware and enterprise-aware AI applications.

A powerful combination

Together, Heroku, Salesforce Data 360, and Agentforce make it possible to:

- Automate the heavy lifting of infrastructure and security requirements to speed up the path from prototype to production for lean teams.
- Unify the entire stack by enabling developers to access all business-critical data and custom logic outside Salesforce CRM.
- Transform agentic AI development from manual infrastructure management to rapid delivery of high-impact AI features without becoming burdened by excessive operational overhead.

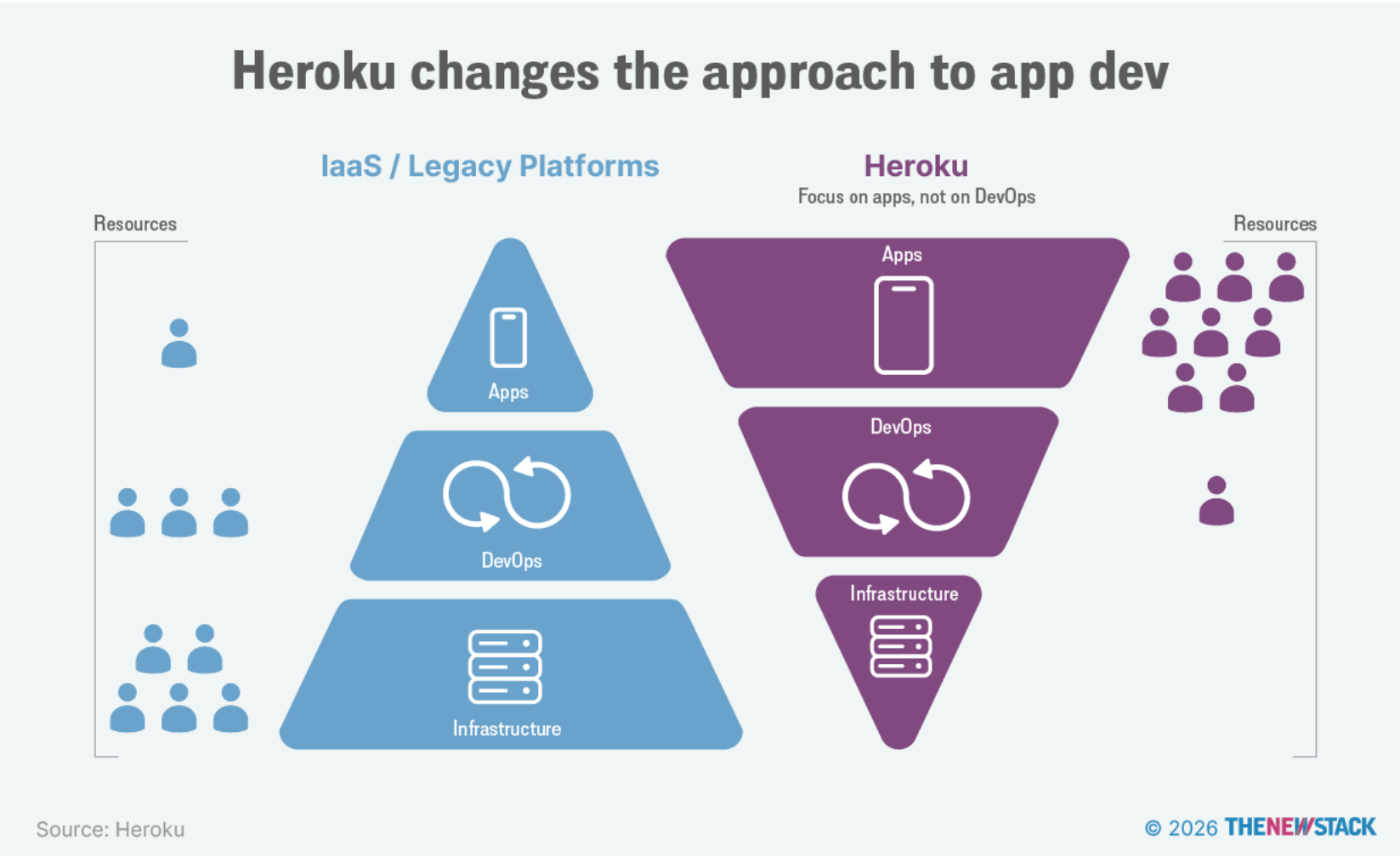


FIG 4.2 By offloading complex compute, infrastructure management, and specialized development to Heroku, developers can deploy more sophisticated, context-aware applications faster and with a leaner operational footprint.

- Simplify compliance and data safety by adding platform-level governance.
- Scale agentic AI actions across the enterprise.

The rise of agentic AI demands a data platform that goes beyond storage to actively empower autonomous AI agents. These agents need to be supplied with intelligent, contextual data from a wide range of data sources (structured or unstructured) in real time and at massive scale. The data used, and the actions taken by AI agents must be unified, harmonized, trustworthy, and accurate, supported by automated data governance, security, and ethical guardrails.

For those operating on the Salesforce Data 360 platform with Agentforce, Heroku extends its services across the enterprise, streamlining development so developers can focus on design rather than bogging down in infrastructure and cybersecurity minutiae.

“The rise of agentic AI demands a data platform that goes beyond storage to actively empower autonomous AI agents.



Conclusion

Most AI projects fail not for lack of ambition but due to friction caused by siloed data, which interferes with developers' efforts to build custom, production-ready infrastructure. Heroku removes this friction by streamlining scale and simplifying DevOps workflows, giving developers the freedom to build with the languages and frameworks they love. As an AI abstraction layer, it provides flexibility to pivot as AI evolves while staying seamlessly integrated with the Salesforce ecosystem.

If agentic AI is to achieve success and scale across the enterprise, it must have real-time access to all CRM and other enterprise data (whether structured or unstructured). Heroku performs this vital function while integrating closely with Salesforce to provide scalability and ease of development across any desired language or platform.

More than a hosting or application platform, Heroku offers the agility required by those designing custom applications and logic that extend Salesforce across the entire enterprise.

By offloading complex compute, infrastructure management, and specialized development to Heroku, developers can deploy more sophisticated, context-aware applications faster and with a leaner operational footprint.

They can connect CRM data to other valuable data and business logic across the enterprise, extending the power of Agentforce and Data 360.

The Heroku developer experience is an app-centric approach for software delivery, integrated with today's most popular developer tools and workflows. It makes it easy to:

- Build in your favorite language
- Streamline app development & deployment

- Reduce operational complexity
- Scale automatically
- Integrate with Salesforce securely and easily

To find out more about how Heroku AI PaaS can help you to deploy and run modern apps, based on a managed container system, with integrated data services and a powerful ecosystem, visit [Heroku: The Cloud Application Platform for Building, Deploying, and Scaling Apps](#)

“Most AI projects fail not for lack of ambition but due to friction caused by siloed data

About the Author



[Drew Robb](#) has been a full-time freelance writer reporting on all areas of IT and energy for more than 25 years.

He has written thousands of articles for *The New Stack*, *Information Week*, *Computerworld*, *eWeek*, *Tech Republic*, *Data Center Knowledge*, *Power Magazine*, *World Trade*, *Renewable Energy World*, and more.

He has a degree in Geology and Geography from the University of Strathclyde UK (USUK). He lives in the Tampa Bay area of Florida. To keep abreast of the IT and energy landscape, he regularly attends industry events such as Data Center World, Mobile World Conference, PowerGen, and others.

Disclosure

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